

WAM CONTACT COUNTERFACTUALS: BRANCH-STRUCTURED PLANNING FOR LATENT CONTACT DYNAMICS

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ABSTRACT

Contact-rich manipulation is not only a prediction problem. A single executed push hides many physically plausible alternatives: the object may be heavier, the contact may slip earlier, or an apparently safe action may enter a different contact mode. This paper studies whether a world-action model for contact-rich manipulation should keep such alternatives explicit until physical observations collapse them. We rebuild WAM Contact Counterfactuals as a real MuJoCo contact-pushing study with latent mass/friction branches, strong MPC baselines, posterior diagnostics, adversarial stress splits, and a frozen evaluation protocol. The proposed Branch-Counterfactual MPC v5 chooses diagnostic actions by bounded expected information gain, updates a posterior over contact branches, and falls back to conservative planning when the latent mode remains underidentified. The paper is intentionally evidence-gated: branch structure is a contribution only if it beats ensemble, adaptive-identification, and robust MPC controls under paired hidden-physics tasks. If the final frozen evidence does not clear that bar, the correct outcome is **STRONG_REVISION**, not a polished but fragile submission claim.

1 INTRODUCTION

Robot manipulation papers often report the trajectory that happened. Contact-rich manipulation also depends on trajectories that could have happened but did not: a pusher might have stuck instead of slipped, a small change in friction might have made the same action underdrive the puck, or a heavier object might have moved just enough during a diagnostic contact to mislead a nominal planner. These unrealized alternatives are not decorative uncertainty. They are action-relevant latent futures.

The central question in this paper is deliberately narrow:

When a robot has a small action budget under hidden contact dynamics, does preserving branch-structured counterfactual futures improve planning beyond averaging or robustifying over the same physical models?

This question matters because the surrounding literature is crowded. Contact-rich manipulation has strong work on diffusion policies, tactile and force-aware world models, visuo-tactile demonstrations, residual correction, and planning-guided policy learning (Seita, 2026; Zheng et al., 2026; Ye et al., 2025; Yu et al., 2025; Chen et al., 2026; Li et al., 2024; Pattabiraman et al., 2024). A paper cannot be novel merely because it adds uncertainty to a planner. It must show that the uncertainty is represented in a way that changes decisions and survives strong controls.

We therefore treat this as a hostile-review exercise rather than a showcase. Earlier versions of this project were too small and too synthetic. The v4 rebuild used real MuJoCo contact dynamics and showed an uncomfortable result: branch MPC beat random and nominal MPC but did not clearly separate from ensemble mean MPC or robust worst-case MPC. The present version expands the method, theory, diagnostics, and final protocol. It also keeps the acceptance gate explicit: if the expanded method still ties strong baselines, the contribution is not ICLR-main-ready.

Contributions. This paper contributes:

1. A formal latent-contact planning setup in which a finite branch library represents physically plausible counterfactual contact futures.
2. A branch-structured MPC method with bounded information-gain diagnostics, posterior reweighting, and entropy-triggered conservative fallback.
3. A negative identifiability analysis explaining when branch structure cannot beat ensemble or robust controls because contact observations do not distinguish latent modes within the action budget.
4. A CPU-only MuJoCo benchmark with paired hidden-physics tasks, strong MPC baselines, adversarial shifts, ablations, posterior diagnostics, bootstrap intervals, paired sign-flip tests, and oracle regret.
5. A submission-readiness decision rule that reports all frozen results honestly and rejects the paper if the mechanism does not beat strong controls.

What this paper does not claim. The method is not a learned visuo-tactile policy, not a hardware result, and not a universal contact manipulation benchmark. It is a controlled mechanism test. That choice is intentional: before scaling to richer perception and hardware, the mechanism must survive the simpler question of whether branch-explicit latent contact planning beats strong planners with access to the same branch library.

2 RELATED WORK AND HOSTILE PRIOR-WORK BOUNDARY

World-action models and contact-rich manipulation. Recent work explores world models, force-aware policies, tactile prediction, and multimodal representations for contact-rich manipulation (Zheng et al., 2026; Ye et al., 2025; Yu et al., 2025; Zang et al., 2026; He et al., 2026). These systems often improve contact performance by adding sensory channels, larger model classes, or richer demonstrations. Our work asks a different mechanistic question: if a planner already has a set of plausible contact dynamics, should it collapse them to an average prediction, guard against the worst branch, identify a maximum-likelihood branch, or preserve branch structure as an object of planning?

Diffusion, VLA, and data augmentation. Diffusion policies and vision-language-action systems are powerful baselines for many manipulation regimes (Li et al., 2024; Seita, 2026; Yu et al., 2025). Data augmentation and demonstration scaling can cover more outcomes, but they do not by themselves specify how a low-budget controller should trade immediate progress against diagnostic contact. The present method can be read as a small, explicit planning module that tests whether counterfactual contact alternatives should be kept separate until an observation discriminates them.

Tactile and force sensing. Tactile and force-aware methods reduce ambiguity by observing contact more directly (Chen et al., 2026; Pattabiraman et al., 2024; Zang et al., 2026). We view such sensing as complementary. In fact, our identifiability analysis predicts that branch-structured planning becomes more valuable when observations separate latent contact modes and less valuable when they do not. A reviewer could fairly ask for tactile hardware validation; this paper does not hide that limitation.

Robust and belief-space planning. Robust MPC protects against poor outcomes by optimizing worst-case or tail-risk objectives (Mayne et al., 2000). POMDP and belief-space planning maintain distributions over latent state (Kaelbling et al., 1998). Our contribution is not the existence of belief updates. The contribution, if supported, is the contact-specific decision structure: a small set of branch rollouts, a diagnostic action constrained by progress regret, posterior diagnostics tied to physical parameters, and an explicit comparison against ensemble, robust, and adaptive-identification MPC using identical MuJoCo tasks.

The hostile novelty boundary. The local hostile prior-work audit identifies the crowded territory as larger models, generic uncertainty, benchmark-only claims, and data augmentation. The defensible boundary is narrower: branch-explicit latent contact alternatives must change closed-loop decisions in a way observed-only or collapsed-belief controls cannot match. If final paired evidence

does not show that, this paper remains a strong empirical scaffold rather than a main-conference submission.

3 LATENT CONTACT PLANNING SETUP

We model a contact-rich manipulation episode as a finite-horizon controlled process with hidden physical mode z . In the MuJoCo benchmark, z corresponds to object mass and contact friction, but the formalism also covers contact-mode support, slip parameters, and unobserved compliance.

Definition 1 (Latent contact planning problem). *A latent contact planning problem is a tuple*

$$\mathcal{P} = (\mathcal{X}, \mathcal{A}, \mathcal{Z}, T_z, O, \ell, H, b_0),$$

where $\mathcal{X}$ is the observed physical state space, $\mathcal{A}$ is an action space, $\mathcal{Z}$ is a latent branch set, $T_z(x'|x, a)$ is the branch-conditioned transition model, $O(o|x, z)$ is an observation model, $\ell(x, a, z)$ is a task loss, H is the action budget, and b_0 is a prior over branches.

At time t , the planner observes o_t , maintains a belief $b_t(z)$, chooses action a_t , observes o_{t+1} , and updates

$$b_{t+1}(z) = \frac{b_t(z)p(o_{t+1} | o_t, a_t, z)}{\sum_{z' \in \mathcal{Z}} b_t(z')p(o_{t+1} | o_t, a_t, z')}.$$

In our implementation, the likelihood is induced by branch rollout error:

$$p(o_{t+1} | o_t, a_t, z) \propto \exp\left(-\frac{\|f_z(o_t, a_t) - o_{t+1}\|_2^2}{2\sigma^2}\right),$$

followed by a small uniform mixing term to avoid irreversible posterior collapse after noisy contacts.

Branch values. For a candidate action a and branch z , let

$$d_z(a) = \|f_z(o_t, a) - g\|_2$$

be the predicted distance to target g . The planners compared in this paper differ in how they aggregate $\{d_z(a)\}_{z \in \mathcal{Z}}$:

$$J_{\text{mean}}(a) = \sum_z b_t(z) d_z(a), \tag{1}$$

$$J_{\text{worst}}(a) = \max_z d_z(a), \tag{2}$$

$$J_{\text{hybrid}}(a) = \alpha J_{\text{mean}}(a) + (1 - \alpha) J_{\text{worst}}(a), \tag{3}$$

$$J_{\text{tail}}(a) = \mathbb{E}_{z \sim b_t} [d_z(a) | d_z(a) \text{ in upper posterior tail}]. \tag{4}$$

Nominal MPC uses a single fixed branch. Ensemble MPC minimizes J_{mean} with a uniform belief. Robust MPC minimizes J_{worst} . Adaptive-ID MPC updates the belief but plans under the maximum-posterior branch. Branch-Counterfactual MPC v5 uses diagnostic actions and then plans with tail or hybrid risk depending on posterior entropy.

4 METHOD

4.1 BRANCH-COUNTERFACTUAL MPC v5

Branch-Counterfactual MPC v5 has three components. First, it keeps a small branch library of plausible physical dynamics. Second, it chooses a bounded diagnostic action that is predicted to reduce branch entropy without sacrificing too much immediate progress. Third, it updates branch weights after contact and falls back to conservative planning when observations do not identify the branch.

Bounded diagnostic action. For each candidate action a , the planner computes branch predictions

$$y_z(a) = f_z(o_t, a).$$

The expected posterior entropy after a is approximated by treating each branch prediction y_z as a possible observation:

$$\hat{H}_{t+1}(a) = \sum_{z \in \mathcal{Z}} b_t(z) H(\text{BayesUpdate}(b_t, a, y_z(a))).$$

The information gain is

$$\text{IG}(a) = H(b_t) - \hat{H}_{t+1}(a).$$

We do not allow arbitrary information-seeking. Let $J_{\min} = \min_{a'} J_{\text{mean}}(a')$. The diagnostic safe set is

$$\mathcal{S}_\epsilon = \{a \in \mathcal{A} : J_{\text{mean}}(a) \leq J_{\min} + \epsilon\}.$$

Branch-Counterfactual MPC v5 selects

$$a^{\text{diag}} = \arg \max_{a \in \mathcal{S}_\epsilon} [\lambda_{\text{ig}} \text{IG}(a) + \lambda_{\text{prog}} \text{Prog}(a) - \lambda_{\text{risk}} J_{\text{mean}}(a)].$$

This construction is intentionally conservative. A diagnostic action must be useful for identification and still remain near the best predicted progress action.

Entropy-triggered fallback. After each contact, the method records posterior entropy. If entropy remains high after a diagnostic contact, the method treats the problem as underidentified and switches to a robust-mean hybrid objective. This is not hidden tuning: fallback activations are reported as a metric. A method that frequently falls back is admitting that the branch evidence did not collapse quickly enough.

Algorithm. For each episode:

1. Initialize a uniform belief over the branch library.
2. On the first push, choose the bounded information-gain diagnostic action.
3. Execute the MuJoCo push under the true hidden mass/friction and any split-specific noise.
4. Update the branch posterior from observed displacement.
5. For remaining pushes, plan with tail risk if entropy is low and robust-mean hybrid risk if entropy is high.
6. Report task metrics and posterior diagnostics.

4.2 BASELINES

The final protocol compares against:

- random push, a weak sanity baseline;
- nominal single-branch MPC;
- ensemble mean MPC, which averages over the same branch library;
- domain-randomized MPC, which replans under sampled branch subsets;
- adaptive-ID MPC, which updates the posterior and then plans under the maximum-posterior branch;
- robust worst-case MPC;
- robust-mean hybrid MPC;
- the legacy v4 branch MPC;
- oracle hidden-parameter MPC, an upper bound that knows the true physical parameters.

These baselines are intentionally uncomfortable. If the proposed method only beats nominal MPC, the branch mechanism is not enough.

5 THEORY: WHEN BRANCHES HELP AND WHEN THEY CANNOT

The theory in this paper is not meant to prove global optimality for contact-rich manipulation. It serves a sharper purpose: identify the conditions under which branch-explicit planning can matter, and the conditions under which no amount of nicer plotting should convert mixed results into a submission claim.

Definition 2 (Branch separation). *For belief b , action a , and branch transition predictions $y_z(a)$, define the posterior-weighted branch separation*

$$S_b(a) = \sum_{z, z'} b(z)b(z') \|y_z(a) - y_{z'}(a)\|_2^2.$$

An action is branch-separating if $S_b(a)$ is large while its task loss remains within the diagnostic safe set.

Proposition 1 (Diagnostic value requires safe separation). *Assume the branch likelihood is Gaussian with variance proxy σ^2 , the second-stage value $V_2(b)$ is L -Lipschitz in total variation distance over beliefs, and there exists an action $a \in \mathcal{S}_\epsilon$ with separation $S_b(a)$. Then the value improvement available to a diagnostic first action is bounded by a term increasing in $S_b(a)/\sigma^2$ and decreasing in the immediate regret ϵ .*

Proof sketch. The Bayes update induced by an observation changes the posterior by an amount controlled by likelihood ratios. Under Gaussian rollout-error likelihoods, expected posterior concentration increases as branch predictions separate relative to σ . The downstream value change is at most Lipschitz in the induced belief change, while the diagnostic safe set caps first-stage regret by ϵ . Thus diagnostic value is positive only when the information-induced second-stage gain exceeds the safe-set regret. $\square$

Proposition 2 (Collapsed baselines are sufficient under action-irrelevant branches). *Suppose all candidate actions induce branch losses whose range is bounded by δ :*

$$\max_{a \in \mathcal{A}} \max_{z, z'} |d_z(a) - d_{z'}(a)| \leq \delta.$$

Then for any belief b , the action chosen by a branch-aware posterior objective can improve over mean-risk planning by at most $O(\delta)$ in predicted one-step loss.

Proof sketch. If every action has nearly the same loss under every branch, then all risk functionals over the branch loss vector are close. Mean, tail, and worst-case scores differ by at most the loss range. Therefore branch structure may still be interpretable, but it is not action-critical. $\square$

Proposition 3 (Underidentification limits branch MPC). *Consider two latent branches z_1, z_2 whose observation distributions after any admissible diagnostic action have total variation distance at most η . After H actions, no policy can distinguish these branches with error probability below a constant controlled by $H\eta$. If the optimal actions for the branches conflict, any non-oracle planner must incur either robust conservatism or branch-selection regret.*

Proof sketch. The result follows from standard testing bounds. If observations under two branches are close in total variation for every action, the transcript distributions induced by any H -step policy remain close up to a factor proportional to $H\eta$. A planner cannot reliably choose branch-specific actions from indistinguishable transcripts. It must either choose a conservative action that performs acceptably on both branches or risk choosing the wrong branch-specific action. $\square$

Corollary 1 (Submission gate). *Branch-explicit planning is a contribution only when the final paired evaluation shows gains over ensemble, adaptive-ID, and robust controls in regimes where branch separation is action-relevant. If those controls match the proposed method, the correct interpretation is not “the method is elegant” but “the mechanism is not yet isolated.”*

6 MUJoCo BENCHMARK

Environment. The benchmark uses direct MuJoCo contact dynamics (Todorov et al., 2012). A spherical pusher moves a cylindrical puck toward a planar target. Hidden physical parameters are puck mass and contact friction. The planner observes puck pose, chooses a small sequence of pushes, and succeeds if the final puck position is within 7.5 cm of the target. The benchmark is deliberately small enough to run on CPU while still using real contact simulation rather than synthetic probability tables.

Branch library. The branch library covers very-low friction, low friction, nominal friction, high friction, light mass, and heavy mass regimes. It is a model class, not a learned perception module. The oracle baseline can use the true hidden parameters, but all non-oracle planners operate through the branch library or nominal branch.

Stress splits. The frozen suite includes nominal, low-friction, high-friction, heavy-object, light-object, observation-noise, actuation-noise, target-shift, contact-offset, combined-shift, and adversarial low-friction/heavy-object splits. Each method receives the same hidden-physics task keys for paired comparisons.

Metrics. Primary metrics are success rate, final distance, and normalized progress. Paired metrics compare Branch-Counterfactual MPC v5 against every baseline on identical split/seed/episode tasks. Diagnostic metrics include posterior entropy, near-true branch mass, branch prediction error, and fallback activation rate. Oracle regret measures how far each non-oracle method remains from hidden-parameter planning.

7 FROZEN EVALUATION PROTOCOL

The final protocol was frozen before running the expanded v5 evaluation. The frozen command is:

```
python src\run_experiment.py --seeds 8 --episodes 20 --workers 2 --chunksize 2
python scripts\analyze_mujoco_results.py
```

This produces 17,600 main rows and 1,440 ablation rows. The protocol uses two CPU workers, single-threaded numeric libraries, and chunked partial writes. The result files are CSV artifacts, not hand-entered numbers. The manuscript imports generated LaTeX tables from those CSVs.

Development versus final evaluation. The method was changed only during development. The final protocol freezes the branch library, splits, baselines, metrics, seeds, episodes, worker count, and decision rule. If a bug is found after the freeze, the fix must be documented and affected final results rerun from scratch.

Decision rule. The paper is ICLR-main-target-ready only if Branch-Counterfactual MPC v5 clearly beats ensemble, domain-randomized, adaptive-ID, robust worst-case, and robust-mean hybrid controls on the predefined primary splits or supports a sharply narrower claim. Ablations must show that information gain, posterior reweighting, and fallback behavior matter. Otherwise the correct terminal decision is **STRONG_REWISE** or **KILL_ARCHIVE**.

8 RESULTS

The final quantitative tables are generated from frozen CSV outputs in Appendix E. We interpret them using three questions.

Q1: Does Branch-Counterfactual MPC v5 beat strong collapsed-belief controls? The central comparison is not against random or nominal MPC. The relevant controls are ensemble mean MPC, adaptive-ID MPC, robust worst-case MPC, and robust-mean hybrid MPC. A positive result requires paired improvements against these baselines, not just higher aggregate means.

Q2: Do diagnostics explain the result? If branch posterior entropy remains high, the theory predicts that the method should struggle or fall back to robust behavior. If near-true branch mass increases and paired performance improves, the diagnostic mechanism is doing work. If success improves without posterior concentration, the claimed mechanism is suspect.

Q3: Do ablations isolate the mechanism? The ablations remove diagnostic pushes, branch reweighting, information gain, conservative fallback, tail risk, and branch-library coverage. The proposed method must beat these variants for the story to be more than an implementation artifact.

9 FAILURE ANALYSIS

The most important failure mode is underidentification. Low-friction contact may produce small or noisy displacement differences across several plausible branches. In that case, a diagnostic action may not collapse the posterior enough before the push budget is exhausted. The method then either falls back to robust planning or risks acting under the wrong branch. This is a scientific result, not an embarrassment to hide. It directly matches Proposition 3.

A second failure mode is robust-baseline saturation. If worst-case or robust-mean planning already selects actions that are good across branch extremes, branch posterior updates may not change the chosen actions. This can make branch planning interpretable but not performance-improving.

A third failure mode is branch-library misspecification. The true mass/friction pair may sit between library points or outside them. Near-true branch mass is therefore a soft diagnostic rather than a guarantee. A learned branch inference module could improve coverage, but that is outside the present controlled study.

10 LIMITATIONS

This paper uses a custom MuJoCo benchmark rather than a public suite or hardware. The branch library is hand-designed. The observation model is puck-pose based rather than tactile-rich. The action set is a compact push stencil, not a learned continuous policy. The strongest modern learned policies are treated as prior-work pressure rather than implemented baselines. These limitations matter. They are why the paper has an explicit readiness gate and why a mixed result should remain **STRONG.REVISE**.

11 REPRODUCIBILITY

The repository contains the runner, analysis script, generated CSVs, figures, protocol freeze, development log, and final status files. The final numbered PDF is copied only to `Downloads/61.pdf`; no numbered paper artifact is placed on the visible Desktop. The CPU-only run uses deterministic task keys derived from split, seed, episode, and method names. Partial CSVs are written after every split and ablation seed so recoverable failures can be resumed or audited.

12 CONCLUSION

WAM Contact Counterfactuals is an intentionally hard-nosed mechanism test. The attractive hypothesis is that contact-rich world-action models should preserve unrealized physical alternatives as branches rather than collapsing them into a single future. The hostile question is whether that representation improves decisions beyond ensemble, adaptive-ID, and robust MPC controls. This paper provides the formal setup, method, theory, benchmark, frozen protocol, and evidence tables needed to answer that question honestly. The final label must follow the evidence.

A DERIVATION DETAILS

A.1 POSTERIOR ENTROPY AND DIAGNOSTIC GAIN

The normalized entropy reported in the experiments is

$$\bar{H}(b) = -\frac{1}{\log |\mathcal{Z}|} \sum_{z \in \mathcal{Z}} b(z) \log b(z).$$

The information-gain diagnostic uses predicted branch observations to approximate expected posterior entropy without sampling new MuJoCo worlds. For action a , the branch prediction matrix provides pairwise errors

$$e_{ij}(a) = \|y_i(a) - y_j(a)\|_2.$$

Treating $y_i(a)$ as the possible observation induced by branch i , the posterior under that observation is

$$b_{t+1}^{(i)}(j) \propto b_t(j) \exp\left(-\frac{e_{ij}(a)^2}{2\sigma^2}\right).$$

The expected entropy is $\sum_i b_t(i) \bar{H}(b_{t+1}^{(i)})$. This approximation is cheap enough for CPU evaluation and directly tied to the rollout predictions already required for planning.

A.2 NEAR-TRUE BRANCH MASS

Because the true hidden mass/friction may not exactly equal a branch-library point, the diagnostic uses a soft kernel:

$$M_{\text{near}}(b, z^*) = \sum_{z_i \in \mathcal{Z}} b(z_i) \exp\left[-\frac{1}{2} \left(\frac{m_i - m^*}{0.08}\right)^2 - \frac{1}{2} \left(\frac{\mu_i - \mu^*}{0.30}\right)^2\right].$$

This metric is not used by the policy. It is reported only for diagnosis because the benchmark knows the hidden physical parameters.

B IMPLEMENTATION NOTES

MuJoCo rollouts. Each branch rollout resets puck and pusher state, executes a position-controlled push, then lets the object settle briefly. Models and data buffers are cached per worker process. This reduces overhead without changing the simulated dynamics.

CPU-light constraints. The final runner uses two process workers and single-threaded numeric libraries. This matters because branch MPC can otherwise multiply CPU usage through candidate actions, branches, and methods. The final branch grid is deliberately small but covers the contact regimes needed for the stress tests.

Why not tune after final results? Tuning after looking at final failures would convert the experiment into an adaptive leaderboard. The protocol freeze prevents that. The paper can still report a failed or mixed result, and the theory section explains why such failures are informative.

C FULL PROTOCOL

The exact protocol is recorded in `docs/paper61_protocol_freeze_20260619.md`. The final suite has:

- 8 seeds;
- 20 episodes per seed/split/method;
- 11 stress splits;
- 10 main methods;
- 9 ablation methods on combined shift;
- 17,600 main rows;
- 1,440 ablation rows.

D REVIEWER ATTACK CHECKLIST

Attack: This is just robust MPC. Response: robust worst-case and robust-mean hybrid MPC are explicit baselines. If they match Branch-Counterfactual MPC v5, the paper does not claim success.

Attack: The branch posterior is decorative. Response: posterior entropy, near-true branch mass, prediction error, fallback activation, and ablations are reported. The mechanism must show up in these diagnostics.

Attack: The benchmark is custom. Response: true. The paper limits its claim to a controlled mechanism test and lists public benchmarks/hardware as required future work for a stronger submission.

Attack: The page count is padded. Response: the appendix includes full frozen protocol tables, paired statistics, per-seed metrics, oracle regret, diagnostics, and derivations generated from artifacts. These are audit evidence, not filler.

E FROZEN QUANTITATIVE RESULTS

These tables are generated directly from the frozen CSV outputs. Empty entries indicate diagnostics that do not apply to a non-adaptive method.

Table 1: Frozen main-suite success and distance for the proposed method and strongest comparators. Entries are mean $\pm$ 95% CI over paired episodes.

Split	Method	Success	Final dist.
act. noise	Adaptive ID	0.431 $\pm$ 0.077	0.086 $\pm$ 0.007
act. noise	Branch v5	0.444 $\pm$ 0.077	0.087 $\pm$ 0.008
act. noise	Ensemble	0.456 $\pm$ 0.077	0.085 $\pm$ 0.007
act. noise	Oracle	0.444 $\pm$ 0.077	0.083 $\pm$ 0.007
act. noise	Robust-mean	0.425 $\pm$ 0.077	0.090 $\pm$ 0.008
act. noise	Robust	0.500 $\pm$ 0.078	0.094 $\pm$ 0.010
adv. low-fric heavy	Adaptive ID	0.025 $\pm$ 0.024	0.161 $\pm$ 0.005
adv. low-fric heavy	Branch v5	0.006 $\pm$ 0.012	0.161 $\pm$ 0.005
adv. low-fric heavy	Ensemble	0.006 $\pm$ 0.012	0.158 $\pm$ 0.005
adv. low-fric heavy	Oracle	0.013 $\pm$ 0.017	0.158 $\pm$ 0.005
adv. low-fric heavy	Robust-mean	0.013 $\pm$ 0.017	0.160 $\pm$ 0.005
adv. low-fric heavy	Robust	0.044 $\pm$ 0.032	0.156 $\pm$ 0.006
combined shift	Adaptive ID	0.419 $\pm$ 0.077	0.096 $\pm$ 0.008
combined shift	Branch v5	0.356 $\pm$ 0.074	0.103 $\pm$ 0.009
combined shift	Ensemble	0.400 $\pm$ 0.076	0.097 $\pm$ 0.009
combined shift	Oracle	0.369 $\pm$ 0.075	0.101 $\pm$ 0.009
combined shift	Robust-mean	0.431 $\pm$ 0.077	0.092 $\pm$ 0.009
combined shift	Robust	0.369 $\pm$ 0.075	0.102 $\pm$ 0.009
contact offset	Adaptive ID	0.231 $\pm$ 0.066	0.133 $\pm$ 0.010
contact offset	Branch v5	0.269 $\pm$ 0.069	0.124 $\pm$ 0.010
contact offset	Ensemble	0.250 $\pm$ 0.067	0.129 $\pm$ 0.010
contact offset	Oracle	0.344 $\pm$ 0.074	0.117 $\pm$ 0.011
contact offset	Robust-mean	0.306 $\pm$ 0.072	0.124 $\pm$ 0.010
contact offset	Robust	0.275 $\pm$ 0.069	0.123 $\pm$ 0.010
heavy object	Adaptive ID	0.444 $\pm$ 0.077	0.078 $\pm$ 0.004
heavy object	Branch v5	0.500 $\pm$ 0.078	0.075 $\pm$ 0.004
heavy object	Ensemble	0.481 $\pm$ 0.078	0.076 $\pm$ 0.004
heavy object	Oracle	0.750 $\pm$ 0.067	0.060 $\pm$ 0.004
heavy object	Robust-mean	0.487 $\pm$ 0.078	0.076 $\pm$ 0.004
heavy object	Robust	0.537 $\pm$ 0.077	0.074 $\pm$ 0.006
high friction	Adaptive ID	0.906 $\pm$ 0.045	0.042 $\pm$ 0.004
high friction	Branch v5	0.787 $\pm$ 0.064	0.053 $\pm$ 0.005
high friction	Ensemble	0.912 $\pm$ 0.044	0.043 $\pm$ 0.004
high friction	Oracle	0.887 $\pm$ 0.049	0.044 $\pm$ 0.004

Split	Method	Success	Final dist.
high friction	Robust-mean	0.931 ± 0.039	0.039 ± 0.004
high friction	Robust	0.963 ± 0.029	0.034 ± 0.003
light object	Adaptive ID	0.556 ± 0.077	0.077 ± 0.005
light object	Branch v5	0.544 ± 0.077	0.080 ± 0.006
light object	Ensemble	0.556 ± 0.077	0.078 ± 0.005
light object	Oracle	0.819 ± 0.060	0.059 ± 0.004
light object	Robust-mean	0.544 ± 0.077	0.079 ± 0.005
light object	Robust	0.594 ± 0.076	0.077 ± 0.007
low friction	Adaptive ID	0.025 ± 0.024	0.107 ± 0.003
low friction	Branch v5	0.169 ± 0.058	0.102 ± 0.005
low friction	Ensemble	0.025 ± 0.024	0.108 ± 0.003
low friction	Oracle	0.544 ± 0.077	0.078 ± 0.006
low friction	Robust-mean	0.025 ± 0.024	0.109 ± 0.003
low friction	Robust	0.231 ± 0.066	0.101 ± 0.007
nominal	Adaptive ID	0.481 ± 0.078	0.078 ± 0.005
nominal	Branch v5	0.463 ± 0.077	0.080 ± 0.005
nominal	Ensemble	0.500 ± 0.078	0.077 ± 0.005
nominal	Oracle	0.762 ± 0.066	0.059 ± 0.004
nominal	Robust-mean	0.481 ± 0.078	0.080 ± 0.006
nominal	Robust	0.606 ± 0.076	0.071 ± 0.007
obs. noise	Adaptive ID	0.438 ± 0.077	0.080 ± 0.005
obs. noise	Branch v5	0.519 ± 0.078	0.075 ± 0.006
obs. noise	Ensemble	0.512 ± 0.078	0.079 ± 0.005
obs. noise	Oracle	0.750 ± 0.067	0.059 ± 0.004
obs. noise	Robust-mean	0.519 ± 0.078	0.077 ± 0.006
obs. noise	Robust	0.531 ± 0.078	0.077 ± 0.007
target shift	Adaptive ID	0.438 ± 0.077	0.086 ± 0.006
target shift	Branch v5	0.450 ± 0.077	0.086 ± 0.006
target shift	Ensemble	0.463 ± 0.077	0.085 ± 0.005
target shift	Oracle	0.606 ± 0.076	0.073 ± 0.005
target shift	Robust-mean	0.469 ± 0.078	0.088 ± 0.006
target shift	Robust	0.550 ± 0.077	0.077 ± 0.008

Table 2: Full frozen main-suite aggregate metrics for every split and method.

Split	Method	Episodes	Success	Final dist.	Progress
act. noise	Adaptive ID	160	0.431 ± 0.077	0.086 ± 0.007	0.726 ± 0.023
act. noise	Branch v4	160	0.544 ± 0.077	0.079 ± 0.007	0.753 ± 0.021
act. noise	Branch v5	160	0.444 ± 0.077	0.087 ± 0.008	0.720 ± 0.028
act. noise	Domain rand.	160	0.431 ± 0.077	0.089 ± 0.007	0.715 ± 0.025
act. noise	Ensemble	160	0.456 ± 0.077	0.085 ± 0.007	0.731 ± 0.023
act. noise	Nominal	160	0.375 ± 0.075	0.089 ± 0.007	0.720 ± 0.024
act. noise	Oracle	160	0.444 ± 0.077	0.083 ± 0.007	0.735 ± 0.023
act. noise	Random	160	0.113 ± 0.049	0.154 ± 0.010	0.509 ± 0.034
act. noise	Robust-mean	160	0.425 ± 0.077	0.090 ± 0.008	0.713 ± 0.027
act. noise	Robust	160	0.500 ± 0.078	0.094 ± 0.010	0.701 ± 0.032
adv. low-fric heavy	Adaptive ID	160	0.025 ± 0.024	0.161 ± 0.005	0.569 ± 0.014
adv. low-fric heavy	Branch v4	160	0.000 ± 0.000	0.155 ± 0.005	0.585 ± 0.014
adv. low-fric heavy	Branch v5	160	0.006 ± 0.012	0.161 ± 0.005	0.568 ± 0.014
adv. low-fric heavy	Domain rand.	160	0.006 ± 0.012	0.157 ± 0.005	0.578 ± 0.015
adv. low-fric heavy	Ensemble	160	0.006 ± 0.012	0.158 ± 0.005	0.574 ± 0.014
adv. low-fric heavy	Nominal	160	0.013 ± 0.017	0.161 ± 0.005	0.565 ± 0.016
adv. low-fric heavy	Oracle	160	0.013 ± 0.017	0.158 ± 0.005	0.576 ± 0.014
adv. low-fric heavy	Random	160	0.000 ± 0.000	0.249 ± 0.018	0.367 ± 0.036
adv. low-fric heavy	Robust-mean	160	0.013 ± 0.017	0.160 ± 0.005	0.573 ± 0.014
adv. low-fric heavy	Robust	160	0.044 ± 0.032	0.156 ± 0.006	0.583 ± 0.016
combined shift	Adaptive ID	160	0.419 ± 0.077	0.096 ± 0.008	0.755 ± 0.021
combined shift	Branch v4	160	0.375 ± 0.075	0.104 ± 0.008	0.732 ± 0.022
combined shift	Branch v5	160	0.356 ± 0.074	0.103 ± 0.009	0.734 ± 0.024
combined shift	Domain rand.	160	0.394 ± 0.076	0.098 ± 0.008	0.747 ± 0.022
combined shift	Ensemble	160	0.400 ± 0.076	0.097 ± 0.009	0.751 ± 0.023

Split	Method	Episodes	Success	Final dist.	Progress
combined shift	Nominal	160	0.269 ± 0.069	0.114 ± 0.009	0.704 ± 0.024
combined shift	Oracle	160	0.369 ± 0.075	0.101 ± 0.009	0.740 ± 0.023
combined shift	Random	160	0.144 ± 0.054	0.170 ± 0.015	0.569 ± 0.035
combined shift	Robust-mean	160	0.431 ± 0.077	0.092 ± 0.009	0.762 ± 0.023
combined shift	Robust	160	0.369 ± 0.075	0.102 ± 0.009	0.737 ± 0.024
contact offset	Adaptive ID	160	0.231 ± 0.066	0.133 ± 0.010	0.586 ± 0.031
contact offset	Branch v4	160	0.263 ± 0.068	0.126 ± 0.010	0.606 ± 0.031
contact offset	Branch v5	160	0.269 ± 0.069	0.124 ± 0.010	0.615 ± 0.032
contact offset	Domain rand.	160	0.306 ± 0.072	0.116 ± 0.009	0.638 ± 0.031
contact offset	Ensemble	160	0.250 ± 0.067	0.129 ± 0.010	0.602 ± 0.033
contact offset	Nominal	160	0.300 ± 0.071	0.121 ± 0.010	0.623 ± 0.032
contact offset	Oracle	160	0.344 ± 0.074	0.117 ± 0.011	0.637 ± 0.033
contact offset	Random	160	0.269 ± 0.069	0.122 ± 0.009	0.621 ± 0.030
contact offset	Robust-mean	160	0.306 ± 0.072	0.124 ± 0.010	0.612 ± 0.033
contact offset	Robust	160	0.275 ± 0.069	0.123 ± 0.010	0.620 ± 0.031
heavy object	Adaptive ID	160	0.444 ± 0.077	0.078 ± 0.004	0.758 ± 0.013
heavy object	Branch v4	160	0.475 ± 0.078	0.076 ± 0.004	0.764 ± 0.013
heavy object	Branch v5	160	0.500 ± 0.078	0.075 ± 0.004	0.765 ± 0.015
heavy object	Domain rand.	160	0.475 ± 0.078	0.075 ± 0.004	0.766 ± 0.013
heavy object	Ensemble	160	0.481 ± 0.078	0.076 ± 0.004	0.764 ± 0.013
heavy object	Nominal	160	0.450 ± 0.077	0.081 ± 0.005	0.750 ± 0.016
heavy object	Oracle	160	0.750 ± 0.067	0.060 ± 0.004	0.815 ± 0.011
heavy object	Random	160	0.087 ± 0.044	0.156 ± 0.010	0.513 ± 0.032
heavy object	Robust-mean	160	0.487 ± 0.078	0.076 ± 0.004	0.761 ± 0.014
heavy object	Robust	160	0.537 ± 0.077	0.074 ± 0.006	0.766 ± 0.019
high friction	Adaptive ID	160	0.906 ± 0.045	0.042 ± 0.004	0.866 ± 0.012
high friction	Branch v4	160	0.912 ± 0.044	0.043 ± 0.004	0.864 ± 0.011
high friction	Branch v5	160	0.787 ± 0.064	0.053 ± 0.005	0.829 ± 0.018
high friction	Domain rand.	160	0.875 ± 0.051	0.045 ± 0.004	0.855 ± 0.014
high friction	Ensemble	160	0.912 ± 0.044	0.043 ± 0.004	0.864 ± 0.011
high friction	Nominal	160	0.694 ± 0.072	0.062 ± 0.006	0.809 ± 0.018
high friction	Oracle	160	0.887 ± 0.049	0.044 ± 0.004	0.862 ± 0.011
high friction	Random	160	0.163 ± 0.057	0.137 ± 0.009	0.559 ± 0.030
high friction	Robust-mean	160	0.931 ± 0.039	0.039 ± 0.004	0.876 ± 0.012
high friction	Robust	160	0.963 ± 0.029	0.034 ± 0.003	0.891 ± 0.011
light object	Adaptive ID	160	0.556 ± 0.077	0.077 ± 0.005	0.754 ± 0.015
light object	Branch v4	160	0.556 ± 0.077	0.078 ± 0.005	0.753 ± 0.014
light object	Branch v5	160	0.544 ± 0.077	0.080 ± 0.006	0.744 ± 0.020
light object	Domain rand.	160	0.537 ± 0.077	0.081 ± 0.005	0.742 ± 0.016
light object	Ensemble	160	0.556 ± 0.077	0.078 ± 0.005	0.753 ± 0.014
light object	Nominal	160	0.569 ± 0.077	0.078 ± 0.005	0.752 ± 0.015
light object	Oracle	160	0.819 ± 0.060	0.059 ± 0.004	0.811 ± 0.012
light object	Random	160	0.094 ± 0.045	0.150 ± 0.009	0.516 ± 0.032
light object	Robust-mean	160	0.544 ± 0.077	0.079 ± 0.005	0.748 ± 0.016
light object	Robust	160	0.594 ± 0.076	0.077 ± 0.007	0.755 ± 0.023
low friction	Adaptive ID	160	0.025 ± 0.024	0.107 ± 0.003	0.662 ± 0.011
low friction	Branch v4	160	0.025 ± 0.024	0.108 ± 0.003	0.661 ± 0.012
low friction	Branch v5	160	0.169 ± 0.058	0.102 ± 0.005	0.687 ± 0.016
low friction	Domain rand.	160	0.056 ± 0.036	0.106 ± 0.004	0.668 ± 0.013
low friction	Ensemble	160	0.025 ± 0.024	0.108 ± 0.003	0.661 ± 0.012
low friction	Nominal	160	0.044 ± 0.032	0.116 ± 0.005	0.637 ± 0.015
low friction	Oracle	160	0.544 ± 0.077	0.078 ± 0.006	0.757 ± 0.017
low friction	Random	160	0.075 ± 0.041	0.183 ± 0.011	0.427 ± 0.036
low friction	Robust-mean	160	0.025 ± 0.024	0.109 ± 0.003	0.658 ± 0.011
low friction	Robust	160	0.231 ± 0.066	0.101 ± 0.007	0.683 ± 0.022
nominal	Adaptive ID	160	0.481 ± 0.078	0.078 ± 0.005	0.756 ± 0.016
nominal	Branch v4	160	0.500 ± 0.078	0.077 ± 0.005	0.758 ± 0.015
nominal	Branch v5	160	0.463 ± 0.077	0.080 ± 0.005	0.746 ± 0.017
nominal	Domain rand.	160	0.500 ± 0.078	0.078 ± 0.005	0.757 ± 0.015
nominal	Ensemble	160	0.500 ± 0.078	0.077 ± 0.005	0.759 ± 0.016
nominal	Nominal	160	0.525 ± 0.078	0.077 ± 0.005	0.758 ± 0.014
nominal	Oracle	160	0.762 ± 0.066	0.059 ± 0.004	0.813 ± 0.013
nominal	Random	160	0.075 ± 0.041	0.160 ± 0.010	0.493 ± 0.033

Split	Method	Episodes	Success	Final dist.	Progress
nominal	Robust-mean	160	0.481 ± 0.078	0.080 ± 0.006	0.748 ± 0.018
nominal	Robust	160	0.606 ± 0.076	0.071 ± 0.007	0.776 ± 0.023
obs. noise	Adaptive ID	160	0.438 ± 0.077	0.080 ± 0.005	0.749 ± 0.017
obs. noise	Branch v4	160	0.537 ± 0.077	0.076 ± 0.005	0.763 ± 0.017
obs. noise	Branch v5	160	0.519 ± 0.078	0.075 ± 0.006	0.764 ± 0.019
obs. noise	Domain rand.	160	0.512 ± 0.078	0.074 ± 0.005	0.765 ± 0.017
obs. noise	Ensemble	160	0.512 ± 0.078	0.079 ± 0.005	0.753 ± 0.015
obs. noise	Nominal	160	0.494 ± 0.078	0.079 ± 0.006	0.754 ± 0.018
obs. noise	Oracle	160	0.750 ± 0.067	0.059 ± 0.004	0.815 ± 0.013
obs. noise	Random	160	0.131 ± 0.052	0.145 ± 0.010	0.542 ± 0.033
obs. noise	Robust-mean	160	0.519 ± 0.078	0.077 ± 0.006	0.761 ± 0.017
obs. noise	Robust	160	0.531 ± 0.078	0.077 ± 0.007	0.758 ± 0.024
target shift	Adaptive ID	160	0.438 ± 0.077	0.086 ± 0.006	0.786 ± 0.013
target shift	Branch v4	160	0.450 ± 0.077	0.085 ± 0.005	0.789 ± 0.013
target shift	Branch v5	160	0.450 ± 0.077	0.086 ± 0.006	0.787 ± 0.013
target shift	Domain rand.	160	0.469 ± 0.078	0.085 ± 0.006	0.790 ± 0.013
target shift	Ensemble	160	0.463 ± 0.077	0.085 ± 0.005	0.790 ± 0.013
target shift	Nominal	160	0.412 ± 0.076	0.090 ± 0.006	0.777 ± 0.014
target shift	Oracle	160	0.606 ± 0.076	0.073 ± 0.005	0.820 ± 0.011
target shift	Random	160	0.125 ± 0.051	0.143 ± 0.010	0.641 ± 0.025
target shift	Robust-mean	160	0.469 ± 0.078	0.088 ± 0.006	0.782 ± 0.015
target shift	Robust	160	0.550 ± 0.077	0.077 ± 0.008	0.809 ± 0.019

Table 3: Paired comparison of Branch v5 against each baseline on identical hidden-physics tasks.

Split	Baseline	Pairs	Δ success	Δ dist.	p(dist.)
act. noise	Random	160	0.331	0.066	0.000
act. noise	Nominal	160	0.069	0.001	0.809
act. noise	Ensemble	160	-0.013	-0.003	0.631
act. noise	Domain rand.	160	0.013	0.002	0.750
act. noise	Adaptive ID	160	0.013	-0.001	0.862
act. noise	Robust	160	-0.056	0.007	0.334
act. noise	Robust-mean	160	0.019	0.002	0.654
act. noise	Branch v4	160	-0.100	-0.009	0.110
act. noise	Oracle	160	0.000	-0.004	0.420
adv. low-fric heavy	Random	160	0.006	0.088	0.000
adv. low-fric heavy	Nominal	160	-0.006	0.000	0.931
adv. low-fric heavy	Ensemble	160	0.000	-0.003	0.289
adv. low-fric heavy	Domain rand.	160	0.000	-0.004	0.100
adv. low-fric heavy	Adaptive ID	160	-0.019	0.000	0.986
adv. low-fric heavy	Robust	160	-0.037	-0.005	0.062
adv. low-fric heavy	Robust-mean	160	-0.006	-0.002	0.574
adv. low-fric heavy	Branch v4	160	0.006	-0.006	0.011
adv. low-fric heavy	Oracle	160	-0.006	-0.003	0.236
combined shift	Random	160	0.212	0.067	0.000
combined shift	Nominal	160	0.087	0.011	0.023
combined shift	Ensemble	160	-0.044	-0.006	0.103
combined shift	Domain rand.	160	-0.037	-0.005	0.228
combined shift	Adaptive ID	160	-0.062	-0.007	0.067
combined shift	Robust	160	-0.013	-0.001	0.800
combined shift	Robust-mean	160	-0.075	-0.011	0.007
combined shift	Branch v4	160	-0.019	0.001	0.875
combined shift	Oracle	160	-0.013	-0.003	0.565
contact offset	Random	160	0.000	-0.002	0.815
contact offset	Nominal	160	-0.031	-0.003	0.655
contact offset	Ensemble	160	0.019	0.005	0.516
contact offset	Domain rand.	160	-0.037	-0.008	0.199
contact offset	Adaptive ID	160	0.037	0.009	0.199
contact offset	Robust	160	-0.006	-0.001	0.891
contact offset	Robust-mean	160	-0.037	0.000	0.976

Split	Baseline	Pairs	Δ success	Δ dist.	p(dist.)
contact offset	Branch v4	160	0.006	0.002	0.797
contact offset	Oracle	160	-0.075	-0.007	0.321
heavy object	Random	160	0.412	0.081	0.000
heavy object	Nominal	160	0.050	0.006	0.027
heavy object	Ensemble	160	0.019	0.001	0.658
heavy object	Domain rand.	160	0.025	0.000	0.987
heavy object	Adaptive ID	160	0.056	0.003	0.206
heavy object	Robust	160	-0.037	-0.001	0.801
heavy object	Robust-mean	160	0.013	0.002	0.403
heavy object	Branch v4	160	0.025	0.001	0.653
heavy object	Oracle	160	-0.250	-0.015	0.000
high friction	Random	160	0.625	0.085	0.000
high friction	Nominal	160	0.094	0.009	0.017
high friction	Ensemble	160	-0.125	-0.010	0.000
high friction	Domain rand.	160	-0.087	-0.007	0.001
high friction	Adaptive ID	160	-0.119	-0.010	0.000
high friction	Robust	160	-0.175	-0.019	0.000
high friction	Robust-mean	160	-0.144	-0.013	0.000
high friction	Branch v4	160	-0.125	-0.010	0.000
high friction	Oracle	160	-0.100	-0.009	0.001
light object	Random	160	0.450	0.070	0.000
light object	Nominal	160	-0.025	-0.002	0.524
light object	Ensemble	160	-0.013	-0.002	0.411
light object	Domain rand.	160	0.006	0.001	0.692
light object	Adaptive ID	160	-0.013	-0.002	0.405
light object	Robust	160	-0.050	-0.003	0.408
light object	Robust-mean	160	0.000	-0.001	0.822
light object	Branch v4	160	-0.013	-0.002	0.378
light object	Oracle	160	-0.275	-0.021	0.000
low friction	Random	160	0.094	0.082	0.000
low friction	Nominal	160	0.125	0.014	0.000
low friction	Ensemble	160	0.144	0.006	0.005
low friction	Domain rand.	160	0.113	0.004	0.015
low friction	Adaptive ID	160	0.144	0.006	0.007
low friction	Robust	160	-0.062	-0.001	0.827
low friction	Robust-mean	160	0.144	0.007	0.002
low friction	Branch v4	160	0.144	0.006	0.003
low friction	Oracle	160	-0.375	-0.024	0.000
nominal	Random	160	0.388	0.080	0.000
nominal	Nominal	160	-0.062	-0.003	0.307
nominal	Ensemble	160	-0.037	-0.003	0.207
nominal	Domain rand.	160	-0.037	-0.002	0.327
nominal	Adaptive ID	160	-0.019	-0.002	0.311
nominal	Robust	160	-0.144	-0.009	0.011
nominal	Robust-mean	160	-0.019	-0.000	0.904
nominal	Branch v4	160	-0.037	-0.003	0.254
nominal	Oracle	160	-0.300	-0.021	0.000
obs. noise	Random	160	0.388	0.070	0.000
obs. noise	Nominal	160	0.025	0.004	0.330
obs. noise	Ensemble	160	0.006	0.004	0.238
obs. noise	Domain rand.	160	0.006	-0.001	0.884
obs. noise	Adaptive ID	160	0.081	0.005	0.169
obs. noise	Robust	160	-0.013	0.002	0.640
obs. noise	Robust-mean	160	0.000	0.002	0.644
obs. noise	Branch v4	160	-0.019	0.001	0.859
obs. noise	Oracle	160	-0.231	-0.016	0.000
target shift	Random	160	0.325	0.057	0.000
target shift	Nominal	160	0.037	0.004	0.137
target shift	Ensemble	160	-0.013	-0.002	0.173
target shift	Domain rand.	160	-0.019	-0.002	0.207
target shift	Adaptive ID	160	0.013	-0.000	0.985
target shift	Robust	160	-0.100	-0.009	0.002
target shift	Robust-mean	160	-0.019	0.002	0.290

Split	Baseline	Pairs	Δ success	Δ dist.	p(dist.)
target shift	Branch v4	160	0.000	-0.001	0.316
target shift	Oracle	160	-0.156	-0.014	0.000

Table 4: Frozen combined-shift ablation results.

Method	Episodes	Success	Final dist.	Progress
Branch v4	160	0.375 ± 0.075	0.104 ± 0.008	0.732 ± 0.022
Branch v5	160	0.356 ± 0.074	0.103 ± 0.009	0.734 ± 0.024
Mean risk	160	0.381 ± 0.075	0.097 ± 0.008	0.748 ± 0.022
No reweighting	160	0.444 ± 0.077	0.096 ± 0.009	0.752 ± 0.024
No fallback	160	0.431 ± 0.077	0.099 ± 0.009	0.747 ± 0.022
No diagnostic	160	0.456 ± 0.077	0.093 ± 0.008	0.761 ± 0.021
No info gain	160	0.412 ± 0.076	0.094 ± 0.009	0.758 ± 0.022
One branch	160	0.319 ± 0.072	0.107 ± 0.009	0.726 ± 0.022
Reduced library	160	0.381 ± 0.075	0.098 ± 0.009	0.751 ± 0.022

Table 5: Posterior and model-diagnostic summaries for adaptive methods.

Split	Method	Entropy	Near-true mass	Pred. err.	Fallbacks
act. noise	Adaptive ID	0.822	0.363	0.054	0.000
act. noise	Branch v4	0.822	0.359	0.053	0.000
act. noise	Branch v5	0.819	0.362	0.059	1.788
adv. low-fric heavy	Adaptive ID	0.564	0.316	0.079	0.000
adv. low-fric heavy	Branch v4	0.550	0.319	0.078	0.000
adv. low-fric heavy	Branch v5	0.560	0.318	0.080	0.544
combined shift	Adaptive ID	0.725	0.270	0.064	0.000
combined shift	Branch v4	0.721	0.264	0.064	0.000
combined shift	Branch v5	0.713	0.262	0.066	1.181
contact offset	Adaptive ID	0.672	0.334	0.150	0.000
contact offset	Branch v4	0.683	0.317	0.149	0.000
contact offset	Branch v5	0.682	0.326	0.149	0.662
heavy object	Adaptive ID	0.850	0.237	0.036	0.000
heavy object	Branch v4	0.850	0.239	0.036	0.000
heavy object	Branch v5	0.835	0.246	0.039	1.919
high friction	Adaptive ID	0.852	0.359	0.039	0.000
high friction	Branch v4	0.851	0.360	0.039	0.000
high friction	Branch v5	0.841	0.363	0.043	1.962
light object	Adaptive ID	0.855	0.322	0.036	0.000
light object	Branch v4	0.856	0.322	0.036	0.000
light object	Branch v5	0.848	0.320	0.039	1.981
low friction	Adaptive ID	0.780	0.438	0.041	0.000
low friction	Branch v4	0.780	0.438	0.041	0.000
low friction	Branch v5	0.756	0.448	0.044	1.438
nominal	Adaptive ID	0.856	0.369	0.035	0.000
nominal	Branch v4	0.856	0.369	0.035	0.000
nominal	Branch v5	0.849	0.373	0.039	1.969
obs. noise	Adaptive ID	0.851	0.362	0.044	0.000
obs. noise	Branch v4	0.846	0.370	0.044	0.000
obs. noise	Branch v5	0.845	0.366	0.046	1.975
target shift	Adaptive ID	0.839	0.377	0.040	0.000
target shift	Branch v4	0.839	0.377	0.041	0.000
target shift	Branch v5	0.835	0.375	0.041	1.850

Table 6: Oracle regret for primary non-oracle methods.

Split	Method	Pairs	Dist. regret	Success regret
act. noise	Adaptive ID	160	0.003	0.013
act. noise	Branch v5	160	0.004	0.000
act. noise	Ensemble	160	0.002	-0.013
act. noise	Robust-mean	160	0.006	0.019
act. noise	Robust	160	0.011	-0.056
adv. low-fric heavy	Adaptive ID	160	0.003	-0.013
adv. low-fric heavy	Branch v5	160	0.003	0.006
adv. low-fric heavy	Ensemble	160	0.001	0.006
adv. low-fric heavy	Robust-mean	160	0.002	0.000
adv. low-fric heavy	Robust	160	-0.002	-0.031
combined shift	Adaptive ID	160	-0.005	-0.050
combined shift	Branch v5	160	0.003	0.013
combined shift	Ensemble	160	-0.004	-0.031
combined shift	Robust-mean	160	-0.008	-0.062
combined shift	Robust	160	0.001	0.000
contact offset	Adaptive ID	160	0.016	0.113
contact offset	Branch v5	160	0.007	0.075
contact offset	Ensemble	160	0.012	0.094
contact offset	Robust-mean	160	0.007	0.037
contact offset	Robust	160	0.006	0.069
heavy object	Adaptive ID	160	0.018	0.306
heavy object	Branch v5	160	0.015	0.250
heavy object	Ensemble	160	0.016	0.269
heavy object	Robust-mean	160	0.017	0.263
heavy object	Robust	160	0.015	0.212
high friction	Adaptive ID	160	-0.002	-0.019
high friction	Branch v5	160	0.009	0.100
high friction	Ensemble	160	-0.001	-0.025
high friction	Robust-mean	160	-0.005	-0.044
high friction	Robust	160	-0.010	-0.075
light object	Adaptive ID	160	0.018	0.263
light object	Branch v5	160	0.021	0.275
light object	Ensemble	160	0.018	0.263
light object	Robust-mean	160	0.020	0.275
light object	Robust	160	0.018	0.225
low friction	Adaptive ID	160	0.029	0.519
low friction	Branch v5	160	0.024	0.375
low friction	Ensemble	160	0.030	0.519
low friction	Robust-mean	160	0.031	0.519
low friction	Robust	160	0.023	0.312
nominal	Adaptive ID	160	0.018	0.281
nominal	Branch v5	160	0.021	0.300
nominal	Ensemble	160	0.018	0.263
nominal	Robust-mean	160	0.020	0.281
nominal	Robust	160	0.012	0.156
obs. noise	Adaptive ID	160	0.020	0.312
obs. noise	Branch v5	160	0.016	0.231
obs. noise	Ensemble	160	0.020	0.237
obs. noise	Robust-mean	160	0.017	0.231
obs. noise	Robust	160	0.018	0.219
target shift	Adaptive ID	160	0.014	0.169
target shift	Branch v5	160	0.014	0.156
target shift	Ensemble	160	0.012	0.144
target shift	Robust-mean	160	0.015	0.138
target shift	Robust	160	0.005	0.056

Table 7: Per-seed frozen metrics for primary methods.

Split	Method	Seed	Success	Final dist.	Progress
act. noise	Adaptive ID	0	0.600	0.082	0.731
act. noise	Adaptive ID	1	0.500	0.094	0.704
act. noise	Adaptive ID	2	0.250	0.103	0.670
act. noise	Adaptive ID	3	0.650	0.068	0.788
act. noise	Adaptive ID	4	0.350	0.091	0.712
act. noise	Adaptive ID	5	0.300	0.087	0.714
act. noise	Adaptive ID	6	0.300	0.091	0.729
act. noise	Adaptive ID	7	0.500	0.075	0.758
act. noise	Branch v5	0	0.300	0.107	0.634
act. noise	Branch v5	1	0.600	0.071	0.776
act. noise	Branch v5	2	0.450	0.087	0.721
act. noise	Branch v5	3	0.350	0.088	0.728
act. noise	Branch v5	4	0.450	0.086	0.732
act. noise	Branch v5	5	0.450	0.082	0.726
act. noise	Branch v5	6	0.350	0.104	0.681
act. noise	Branch v5	7	0.600	0.075	0.761
act. noise	Ensemble	0	0.300	0.095	0.677
act. noise	Ensemble	1	0.400	0.086	0.736
act. noise	Ensemble	2	0.500	0.083	0.738
act. noise	Ensemble	3	0.500	0.078	0.757
act. noise	Ensemble	4	0.650	0.073	0.771
act. noise	Ensemble	5	0.350	0.090	0.704
act. noise	Ensemble	6	0.500	0.082	0.752
act. noise	Ensemble	7	0.450	0.090	0.711
act. noise	Oracle	0	0.500	0.076	0.738
act. noise	Oracle	1	0.550	0.083	0.738
act. noise	Oracle	2	0.500	0.084	0.731
act. noise	Oracle	3	0.400	0.089	0.731
act. noise	Oracle	4	0.350	0.087	0.729
act. noise	Oracle	5	0.300	0.083	0.728
act. noise	Oracle	6	0.500	0.072	0.779
act. noise	Oracle	7	0.450	0.091	0.706
act. noise	Robust-mean	0	0.300	0.095	0.681
act. noise	Robust-mean	1	0.550	0.086	0.721
act. noise	Robust-mean	2	0.500	0.088	0.721
act. noise	Robust-mean	3	0.350	0.101	0.685
act. noise	Robust-mean	4	0.450	0.076	0.761
act. noise	Robust-mean	5	0.300	0.098	0.675
act. noise	Robust-mean	6	0.500	0.083	0.751
act. noise	Robust-mean	7	0.450	0.091	0.710
act. noise	Robust	0	0.500	0.077	0.748
act. noise	Robust	1	0.450	0.115	0.636
act. noise	Robust	2	0.400	0.100	0.673
act. noise	Robust	3	0.400	0.118	0.621
act. noise	Robust	4	0.650	0.071	0.786
act. noise	Robust	5	0.450	0.104	0.662
act. noise	Robust	6	0.500	0.087	0.738
act. noise	Robust	7	0.650	0.081	0.746
adv. low-fric heavy	Adaptive ID	0	0.050	0.150	0.607
adv. low-fric heavy	Adaptive ID	1	0.100	0.152	0.586
adv. low-fric heavy	Adaptive ID	2	0.000	0.167	0.553
adv. low-fric heavy	Adaptive ID	3	0.000	0.156	0.586
adv. low-fric heavy	Adaptive ID	4	0.000	0.176	0.538
adv. low-fric heavy	Adaptive ID	5	0.000	0.166	0.557
adv. low-fric heavy	Adaptive ID	6	0.000	0.155	0.558
adv. low-fric heavy	Adaptive ID	7	0.050	0.166	0.563
adv. low-fric heavy	Branch v5	0	0.000	0.159	0.584
adv. low-fric heavy	Branch v5	1	0.000	0.164	0.553
adv. low-fric heavy	Branch v5	2	0.050	0.158	0.573
adv. low-fric heavy	Branch v5	3	0.000	0.161	0.572
adv. low-fric heavy	Branch v5	4	0.000	0.164	0.567

Split	Method	Seed	Success	Final dist.	Progress
adv. low-fric heavy	Branch v5	5	0.000	0.159	0.574
adv. low-fric heavy	Branch v5	6	0.000	0.155	0.562
adv. low-fric heavy	Branch v5	7	0.000	0.168	0.557
adv. low-fric heavy	Ensemble	0	0.000	0.144	0.619
adv. low-fric heavy	Ensemble	1	0.050	0.163	0.554
adv. low-fric heavy	Ensemble	2	0.000	0.161	0.567
adv. low-fric heavy	Ensemble	3	0.000	0.157	0.584
adv. low-fric heavy	Ensemble	4	0.000	0.167	0.556
adv. low-fric heavy	Ensemble	5	0.000	0.156	0.584
adv. low-fric heavy	Ensemble	6	0.000	0.153	0.568
adv. low-fric heavy	Ensemble	7	0.000	0.167	0.560
adv. low-fric heavy	Oracle	0	0.000	0.150	0.609
adv. low-fric heavy	Oracle	1	0.000	0.148	0.593
adv. low-fric heavy	Oracle	2	0.000	0.168	0.552
adv. low-fric heavy	Oracle	3	0.050	0.150	0.601
adv. low-fric heavy	Oracle	4	0.000	0.169	0.553
adv. low-fric heavy	Oracle	5	0.000	0.164	0.560
adv. low-fric heavy	Oracle	6	0.050	0.154	0.567
adv. low-fric heavy	Oracle	7	0.000	0.162	0.577
adv. low-fric heavy	Robust-mean	0	0.000	0.158	0.587
adv. low-fric heavy	Robust-mean	1	0.050	0.162	0.553
adv. low-fric heavy	Robust-mean	2	0.000	0.161	0.569
adv. low-fric heavy	Robust-mean	3	0.000	0.152	0.597
adv. low-fric heavy	Robust-mean	4	0.000	0.164	0.566
adv. low-fric heavy	Robust-mean	5	0.000	0.164	0.569
adv. low-fric heavy	Robust-mean	6	0.050	0.149	0.578
adv. low-fric heavy	Robust-mean	7	0.000	0.166	0.563
adv. low-fric heavy	Robust	0	0.100	0.147	0.614
adv. low-fric heavy	Robust	1	0.150	0.144	0.611
adv. low-fric heavy	Robust	2	0.000	0.156	0.580
adv. low-fric heavy	Robust	3	0.000	0.164	0.564
adv. low-fric heavy	Robust	4	0.050	0.163	0.569
adv. low-fric heavy	Robust	5	0.000	0.159	0.578
adv. low-fric heavy	Robust	6	0.050	0.146	0.589
adv. low-fric heavy	Robust	7	0.000	0.169	0.556
combined shift	Adaptive ID	0	0.350	0.105	0.733
combined shift	Adaptive ID	1	0.450	0.095	0.759
combined shift	Adaptive ID	2	0.500	0.081	0.785
combined shift	Adaptive ID	3	0.400	0.088	0.769
combined shift	Adaptive ID	4	0.450	0.108	0.719
combined shift	Adaptive ID	5	0.400	0.097	0.764
combined shift	Adaptive ID	6	0.500	0.090	0.767
combined shift	Adaptive ID	7	0.300	0.102	0.742
combined shift	Branch v5	0	0.300	0.113	0.711
combined shift	Branch v5	1	0.350	0.099	0.745
combined shift	Branch v5	2	0.350	0.092	0.759
combined shift	Branch v5	3	0.400	0.097	0.745
combined shift	Branch v5	4	0.250	0.106	0.730
combined shift	Branch v5	5	0.350	0.108	0.729
combined shift	Branch v5	6	0.400	0.107	0.717
combined shift	Branch v5	7	0.450	0.103	0.738
combined shift	Ensemble	0	0.200	0.116	0.703
combined shift	Ensemble	1	0.450	0.084	0.785
combined shift	Ensemble	2	0.350	0.102	0.732
combined shift	Ensemble	3	0.550	0.093	0.754
combined shift	Ensemble	4	0.500	0.086	0.777
combined shift	Ensemble	5	0.400	0.094	0.769
combined shift	Ensemble	6	0.350	0.103	0.730
combined shift	Ensemble	7	0.400	0.095	0.756
combined shift	Oracle	0	0.350	0.108	0.727
combined shift	Oracle	1	0.500	0.086	0.781
combined shift	Oracle	2	0.300	0.101	0.738
combined shift	Oracle	3	0.350	0.094	0.751

Split	Method	Seed	Success	Final dist.	Progress
combined shift	Oracle	4	0.300	0.112	0.711
combined shift	Oracle	5	0.500	0.091	0.771
combined shift	Oracle	6	0.300	0.111	0.699
combined shift	Oracle	7	0.350	0.101	0.738
combined shift	Robust-mean	0	0.400	0.095	0.762
combined shift	Robust-mean	1	0.400	0.099	0.752
combined shift	Robust-mean	2	0.500	0.078	0.791
combined shift	Robust-mean	3	0.550	0.087	0.776
combined shift	Robust-mean	4	0.450	0.093	0.751
combined shift	Robust-mean	5	0.400	0.096	0.764
combined shift	Robust-mean	6	0.350	0.098	0.741
combined shift	Robust-mean	7	0.400	0.093	0.758
combined shift	Robust	0	0.300	0.114	0.704
combined shift	Robust	1	0.350	0.104	0.740
combined shift	Robust	2	0.250	0.106	0.729
combined shift	Robust	3	0.400	0.094	0.751
combined shift	Robust	4	0.300	0.114	0.704
combined shift	Robust	5	0.450	0.099	0.758
combined shift	Robust	6	0.400	0.096	0.743
combined shift	Robust	7	0.500	0.089	0.765
contact offset	Adaptive ID	0	0.350	0.136	0.590
contact offset	Adaptive ID	1	0.150	0.134	0.570
contact offset	Adaptive ID	2	0.200	0.131	0.578
contact offset	Adaptive ID	3	0.300	0.123	0.594
contact offset	Adaptive ID	4	0.300	0.120	0.622
contact offset	Adaptive ID	5	0.200	0.138	0.576
contact offset	Adaptive ID	6	0.150	0.139	0.595
contact offset	Adaptive ID	7	0.200	0.143	0.560
contact offset	Branch v5	0	0.350	0.108	0.653
contact offset	Branch v5	1	0.350	0.104	0.662
contact offset	Branch v5	2	0.350	0.113	0.647
contact offset	Branch v5	3	0.300	0.129	0.594
contact offset	Branch v5	4	0.300	0.129	0.587
contact offset	Branch v5	5	0.350	0.110	0.667
contact offset	Branch v5	6	0.000	0.167	0.513
contact offset	Branch v5	7	0.150	0.131	0.595
contact offset	Ensemble	0	0.450	0.120	0.638
contact offset	Ensemble	1	0.250	0.127	0.600
contact offset	Ensemble	2	0.150	0.142	0.544
contact offset	Ensemble	3	0.100	0.125	0.605
contact offset	Ensemble	4	0.250	0.141	0.549
contact offset	Ensemble	5	0.100	0.147	0.549
contact offset	Ensemble	6	0.150	0.141	0.588
contact offset	Ensemble	7	0.550	0.085	0.739
contact offset	Oracle	0	0.300	0.114	0.649
contact offset	Oracle	1	0.500	0.097	0.679
contact offset	Oracle	2	0.350	0.106	0.664
contact offset	Oracle	3	0.300	0.135	0.575
contact offset	Oracle	4	0.300	0.119	0.623
contact offset	Oracle	5	0.250	0.137	0.582
contact offset	Oracle	6	0.450	0.093	0.735
contact offset	Oracle	7	0.300	0.135	0.586
contact offset	Robust-mean	0	0.350	0.108	0.662
contact offset	Robust-mean	1	0.250	0.123	0.606
contact offset	Robust-mean	2	0.350	0.125	0.602
contact offset	Robust-mean	3	0.400	0.110	0.637
contact offset	Robust-mean	4	0.250	0.134	0.577
contact offset	Robust-mean	5	0.250	0.131	0.593
contact offset	Robust-mean	6	0.200	0.138	0.603
contact offset	Robust-mean	7	0.400	0.124	0.621
contact offset	Robust	0	0.350	0.120	0.640
contact offset	Robust	1	0.300	0.120	0.608
contact offset	Robust	2	0.300	0.128	0.607

Split	Method	Seed	Success	Final dist.	Progress
contact offset	Robust	3	0.250	0.126	0.603
contact offset	Robust	4	0.250	0.123	0.598
contact offset	Robust	5	0.250	0.126	0.626
contact offset	Robust	6	0.150	0.122	0.642
contact offset	Robust	7	0.350	0.118	0.633
heavy object	Adaptive ID	0	0.350	0.079	0.750
heavy object	Adaptive ID	1	0.350	0.084	0.744
heavy object	Adaptive ID	2	0.350	0.082	0.734
heavy object	Adaptive ID	3	0.500	0.073	0.766
heavy object	Adaptive ID	4	0.450	0.082	0.741
heavy object	Adaptive ID	5	0.600	0.073	0.768
heavy object	Adaptive ID	6	0.500	0.075	0.778
heavy object	Adaptive ID	7	0.450	0.073	0.784
heavy object	Branch v5	0	0.550	0.070	0.784
heavy object	Branch v5	1	0.350	0.081	0.752
heavy object	Branch v5	2	0.450	0.078	0.749
heavy object	Branch v5	3	0.550	0.075	0.753
heavy object	Branch v5	4	0.550	0.075	0.762
heavy object	Branch v5	5	0.550	0.078	0.747
heavy object	Branch v5	6	0.500	0.071	0.791
heavy object	Branch v5	7	0.500	0.070	0.787
heavy object	Ensemble	0	0.450	0.074	0.769
heavy object	Ensemble	1	0.350	0.083	0.748
heavy object	Ensemble	2	0.350	0.084	0.729
heavy object	Ensemble	3	0.550	0.071	0.771
heavy object	Ensemble	4	0.450	0.081	0.742
heavy object	Ensemble	5	0.650	0.073	0.765
heavy object	Ensemble	6	0.550	0.071	0.789
heavy object	Ensemble	7	0.500	0.069	0.795
heavy object	Oracle	0	0.700	0.064	0.803
heavy object	Oracle	1	0.650	0.066	0.800
heavy object	Oracle	2	0.850	0.056	0.820
heavy object	Oracle	3	0.750	0.059	0.809
heavy object	Oracle	4	0.850	0.053	0.833
heavy object	Oracle	5	0.750	0.065	0.796
heavy object	Oracle	6	0.700	0.053	0.846
heavy object	Oracle	7	0.750	0.063	0.815
heavy object	Robust-mean	0	0.500	0.072	0.774
heavy object	Robust-mean	1	0.350	0.086	0.738
heavy object	Robust-mean	2	0.350	0.082	0.735
heavy object	Robust-mean	3	0.550	0.074	0.757
heavy object	Robust-mean	4	0.450	0.082	0.741
heavy object	Robust-mean	5	0.650	0.074	0.762
heavy object	Robust-mean	6	0.550	0.071	0.793
heavy object	Robust-mean	7	0.500	0.071	0.788
heavy object	Robust	0	0.550	0.076	0.758
heavy object	Robust	1	0.400	0.077	0.761
heavy object	Robust	2	0.350	0.085	0.718
heavy object	Robust	3	0.550	0.079	0.741
heavy object	Robust	4	0.600	0.065	0.789
heavy object	Robust	5	0.650	0.070	0.774
heavy object	Robust	6	0.700	0.063	0.815
heavy object	Robust	7	0.500	0.077	0.769
high friction	Adaptive ID	0	0.950	0.042	0.866
high friction	Adaptive ID	1	0.900	0.043	0.867
high friction	Adaptive ID	2	0.950	0.037	0.887
high friction	Adaptive ID	3	0.850	0.046	0.848
high friction	Adaptive ID	4	0.950	0.036	0.885
high friction	Adaptive ID	5	0.800	0.049	0.849
high friction	Adaptive ID	6	0.900	0.039	0.869
high friction	Adaptive ID	7	0.950	0.047	0.854
high friction	Branch v5	0	0.800	0.054	0.823
high friction	Branch v5	1	0.800	0.049	0.844

	Split	Method	Seed	Success	Final dist.	Progress
1026						
1027						
1028	high friction	Branch v5	2	0.900	0.044	0.864
1029	high friction	Branch v5	3	0.550	0.068	0.765
1030	high friction	Branch v5	4	0.900	0.043	0.857
1031	high friction	Branch v5	5	0.600	0.065	0.792
1032	high friction	Branch v5	6	0.850	0.048	0.835
1033	high friction	Branch v5	7	0.900	0.049	0.849
1034	high friction	Ensemble	0	0.950	0.043	0.863
1035	high friction	Ensemble	1	0.900	0.043	0.867
1036	high friction	Ensemble	2	0.950	0.039	0.882
1037	high friction	Ensemble	3	0.850	0.046	0.845
1038	high friction	Ensemble	4	0.950	0.038	0.879
1039	high friction	Ensemble	5	0.800	0.049	0.848
1040	high friction	Ensemble	6	0.950	0.038	0.871
1041	high friction	Ensemble	7	0.950	0.046	0.856
1042	high friction	Oracle	0	0.900	0.049	0.844
1043	high friction	Oracle	1	0.750	0.045	0.864
1044	high friction	Oracle	2	0.900	0.045	0.865
1045	high friction	Oracle	3	0.800	0.046	0.847
1046	high friction	Oracle	4	0.950	0.042	0.865
1047	high friction	Oracle	5	0.800	0.048	0.852
1048	high friction	Oracle	6	1.000	0.033	0.889
1049	high friction	Oracle	7	1.000	0.043	0.868
1050	high friction	Robust-mean	0	0.950	0.041	0.868
1051	high friction	Robust-mean	1	0.900	0.041	0.875
1052	high friction	Robust-mean	2	0.950	0.035	0.897
1053	high friction	Robust-mean	3	0.850	0.043	0.861
1054	high friction	Robust-mean	4	1.000	0.034	0.892
1055	high friction	Robust-mean	5	0.900	0.040	0.875
1056	high friction	Robust-mean	6	0.950	0.037	0.874
1057	high friction	Robust-mean	7	0.950	0.045	0.862
1058	high friction	Robust	0	0.950	0.038	0.880
1059	high friction	Robust	1	0.950	0.031	0.902
1060	high friction	Robust	2	0.950	0.033	0.902
1061	high friction	Robust	3	0.950	0.033	0.887
1062	high friction	Robust	4	1.000	0.034	0.892
1063	high friction	Robust	5	0.950	0.034	0.891
1064	high friction	Robust	6	1.000	0.035	0.879
1065	high friction	Robust	7	0.950	0.035	0.893
1066	light object	Adaptive ID	0	0.650	0.077	0.754
1067	light object	Adaptive ID	1	0.450	0.082	0.757
1068	light object	Adaptive ID	2	0.700	0.072	0.759
1069	light object	Adaptive ID	3	0.400	0.080	0.743
1070	light object	Adaptive ID	4	0.550	0.078	0.749
1071	light object	Adaptive ID	5	0.550	0.080	0.744
1072	light object	Adaptive ID	6	0.500	0.079	0.765
1073	light object	Adaptive ID	7	0.650	0.073	0.762
1074	light object	Branch v5	0	0.550	0.079	0.743
1075	light object	Branch v5	1	0.400	0.090	0.727
1076	light object	Branch v5	2	0.700	0.072	0.758
1077	light object	Branch v5	3	0.500	0.083	0.729
1078	light object	Branch v5	4	0.550	0.079	0.742
1079	light object	Branch v5	5	0.550	0.084	0.733
	light object	Branch v5	6	0.500	0.084	0.737
	light object	Branch v5	7	0.600	0.068	0.784
	light object	Ensemble	0	0.650	0.077	0.754
	light object	Ensemble	1	0.450	0.083	0.753
	light object	Ensemble	2	0.700	0.073	0.756
	light object	Ensemble	3	0.400	0.083	0.729
	light object	Ensemble	4	0.600	0.075	0.757
	light object	Ensemble	5	0.550	0.080	0.742
	light object	Ensemble	6	0.500	0.075	0.774
	light object	Ensemble	7	0.600	0.075	0.757
	light object	Oracle	0	0.850	0.055	0.820

	Split	Method	Seed	Success	Final dist.	Progress
1080						
1081						
1082	light object	Oracle	1	0.750	0.063	0.813
1083	light object	Oracle	2	0.900	0.056	0.811
1084	light object	Oracle	3	0.650	0.069	0.777
1085	light object	Oracle	4	0.950	0.052	0.831
1086	light object	Oracle	5	0.750	0.059	0.814
1087	light object	Oracle	6	0.850	0.060	0.820
1088	light object	Oracle	7	0.850	0.061	0.803
1089	light object	Robust-mean	0	0.600	0.080	0.742
1090	light object	Robust-mean	1	0.450	0.084	0.750
1091	light object	Robust-mean	2	0.700	0.072	0.758
1092	light object	Robust-mean	3	0.400	0.082	0.736
1093	light object	Robust-mean	4	0.600	0.074	0.760
1094	light object	Robust-mean	5	0.500	0.087	0.722
1095	light object	Robust-mean	6	0.500	0.077	0.768
1096	light object	Robust-mean	7	0.600	0.077	0.752
1097	light object	Robust	0	0.650	0.083	0.736
1098	light object	Robust	1	0.550	0.083	0.756
1099	light object	Robust	2	0.850	0.064	0.779
1100	light object	Robust	3	0.450	0.078	0.748
1101	light object	Robust	4	0.550	0.085	0.725
1102	light object	Robust	5	0.450	0.087	0.725
1103	light object	Robust	6	0.700	0.056	0.831
1104	light object	Robust	7	0.550	0.079	0.743
1105	low friction	Adaptive ID	0	0.000	0.112	0.650
1106	low friction	Adaptive ID	1	0.000	0.112	0.630
1107	low friction	Adaptive ID	2	0.050	0.106	0.670
1108	low friction	Adaptive ID	3	0.000	0.109	0.673
1109	low friction	Adaptive ID	4	0.050	0.103	0.662
1110	low friction	Adaptive ID	5	0.050	0.099	0.693
1111	low friction	Adaptive ID	6	0.050	0.107	0.671
1112	low friction	Adaptive ID	7	0.000	0.112	0.648
1113	low friction	Branch v5	0	0.200	0.104	0.684
1114	low friction	Branch v5	1	0.200	0.102	0.670
1115	low friction	Branch v5	2	0.200	0.094	0.717
1116	low friction	Branch v5	3	0.150	0.107	0.680
1117	low friction	Branch v5	4	0.150	0.101	0.674
1118	low friction	Branch v5	5	0.150	0.095	0.709
1119	low friction	Branch v5	6	0.200	0.105	0.685
1120	low friction	Branch v5	7	0.100	0.105	0.673
1121	low friction	Ensemble	0	0.000	0.115	0.638
1122	low friction	Ensemble	1	0.000	0.112	0.630
1123	low friction	Ensemble	2	0.050	0.106	0.670
1124	low friction	Ensemble	3	0.000	0.111	0.669
1125	low friction	Ensemble	4	0.050	0.101	0.671
1126	low friction	Ensemble	5	0.050	0.100	0.692
1127	low friction	Ensemble	6	0.050	0.107	0.671
1128	low friction	Ensemble	7	0.000	0.112	0.648
1129	low friction	Oracle	0	0.500	0.085	0.737
1130	low friction	Oracle	1	0.450	0.082	0.727
1131	low friction	Oracle	2	0.550	0.076	0.771
1132	low friction	Oracle	3	0.550	0.084	0.747
1133	low friction	Oracle	4	0.550	0.074	0.754
	low friction	Oracle	5	0.650	0.064	0.809
	low friction	Oracle	6	0.550	0.079	0.761
	low friction	Oracle	7	0.550	0.081	0.751
	low friction	Robust-mean	0	0.000	0.114	0.643
	low friction	Robust-mean	1	0.000	0.110	0.638
	low friction	Robust-mean	2	0.050	0.108	0.663
	low friction	Robust-mean	3	0.000	0.112	0.663
	low friction	Robust-mean	4	0.050	0.106	0.652
	low friction	Robust-mean	5	0.050	0.101	0.687
	low friction	Robust-mean	6	0.050	0.109	0.665
	low friction	Robust-mean	7	0.000	0.112	0.648

	Split	Method	Seed	Success	Final dist.	Progress
1134						
1135						
1136	low friction	Robust	0	0.100	0.114	0.644
1137	low friction	Robust	1	0.350	0.095	0.686
1138	low friction	Robust	2	0.200	0.107	0.661
1139	low friction	Robust	3	0.300	0.093	0.725
1140	low friction	Robust	4	0.150	0.102	0.660
1141	low friction	Robust	5	0.300	0.086	0.732
1142	low friction	Robust	6	0.150	0.106	0.677
1143	nominal	Adaptive ID	7	0.300	0.103	0.681
1144	nominal	Adaptive ID	0	0.400	0.078	0.758
1145	nominal	Adaptive ID	1	0.300	0.084	0.736
1146	nominal	Adaptive ID	2	0.700	0.071	0.767
1147	nominal	Adaptive ID	3	0.450	0.079	0.752
1148	nominal	Adaptive ID	4	0.400	0.079	0.753
1149	nominal	Adaptive ID	5	0.600	0.074	0.759
1150	nominal	Adaptive ID	6	0.450	0.079	0.758
1151	nominal	Adaptive ID	7	0.550	0.077	0.763
1152	nominal	Branch v5	0	0.400	0.083	0.736
1153	nominal	Branch v5	1	0.500	0.076	0.767
1154	nominal	Branch v5	2	0.600	0.075	0.748
1155	nominal	Branch v5	3	0.350	0.081	0.743
1156	nominal	Branch v5	4	0.350	0.089	0.721
1157	nominal	Branch v5	5	0.600	0.078	0.743
1158	nominal	Branch v5	6	0.400	0.081	0.753
1159	nominal	Branch v5	7	0.500	0.077	0.761
1160	nominal	Ensemble	0	0.450	0.074	0.770
1161	nominal	Ensemble	1	0.350	0.083	0.742
1162	nominal	Ensemble	2	0.750	0.069	0.778
1163	nominal	Ensemble	3	0.400	0.083	0.738
1164	nominal	Ensemble	4	0.400	0.081	0.751
1165	nominal	Ensemble	5	0.650	0.071	0.767
1166	nominal	Ensemble	6	0.450	0.078	0.760
1167	nominal	Ensemble	7	0.550	0.076	0.769
1168	nominal	Oracle	0	0.800	0.058	0.815
1169	nominal	Oracle	1	0.800	0.055	0.829
1170	nominal	Oracle	2	0.950	0.056	0.817
1171	nominal	Oracle	3	0.700	0.063	0.801
1172	nominal	Oracle	4	0.650	0.061	0.810
1173	nominal	Oracle	5	0.900	0.054	0.823
1174	nominal	Oracle	6	0.700	0.056	0.828
1175	nominal	Oracle	7	0.600	0.072	0.780
1176	nominal	Robust-mean	0	0.450	0.077	0.754
1177	nominal	Robust-mean	1	0.350	0.083	0.742
1178	nominal	Robust-mean	2	0.700	0.067	0.783
1179	nominal	Robust-mean	3	0.400	0.085	0.734
1180	nominal	Robust-mean	4	0.400	0.082	0.747
1181	nominal	Robust-mean	5	0.600	0.075	0.750
1182	nominal	Robust-mean	6	0.450	0.085	0.741
1183	nominal	Robust-mean	7	0.500	0.085	0.736
1184	nominal	Robust	0	0.500	0.073	0.770
1185	nominal	Robust	1	0.550	0.080	0.751
1186	nominal	Robust	2	0.750	0.058	0.812
1187	nominal	Robust	3	0.600	0.073	0.779
1188	nominal	Robust	4	0.650	0.068	0.784
1189	nominal	Robust	5	0.650	0.066	0.781
1190	nominal	Robust	6	0.550	0.079	0.759
1191	nominal	Robust	7	0.600	0.074	0.773
1192	obs. noise	Adaptive ID	0	0.400	0.081	0.735
1193	obs. noise	Adaptive ID	1	0.450	0.071	0.767
1194	obs. noise	Adaptive ID	2	0.250	0.095	0.717
1195	obs. noise	Adaptive ID	3	0.600	0.074	0.768
1196	obs. noise	Adaptive ID	4	0.650	0.063	0.809
1197	obs. noise	Adaptive ID	5	0.300	0.095	0.680
1198	obs. noise	Adaptive ID	6	0.500	0.077	0.770

	Split	Method	Seed	Success	Final dist.	Progress
1188						
1189						
1190	obs. noise	Adaptive ID	7	0.350	0.081	0.742
1191	obs. noise	Branch v5	0	0.450	0.077	0.756
1192	obs. noise	Branch v5	1	0.550	0.073	0.758
1193	obs. noise	Branch v5	2	0.550	0.077	0.772
1194	obs. noise	Branch v5	3	0.650	0.071	0.783
1195	obs. noise	Branch v5	4	0.450	0.078	0.755
1196	obs. noise	Branch v5	5	0.350	0.086	0.711
1197	obs. noise	Branch v5	6	0.700	0.059	0.825
1198	obs. noise	Branch v5	7	0.450	0.078	0.753
1199	obs. noise	Ensemble	0	0.600	0.074	0.763
1200	obs. noise	Ensemble	1	0.600	0.074	0.762
1201	obs. noise	Ensemble	2	0.500	0.074	0.780
1202	obs. noise	Ensemble	3	0.450	0.080	0.754
1203	obs. noise	Ensemble	4	0.350	0.090	0.724
1204	obs. noise	Ensemble	5	0.600	0.073	0.760
1205	obs. noise	Ensemble	6	0.450	0.082	0.753
1206	obs. noise	Ensemble	7	0.550	0.087	0.727
1207	obs. noise	Oracle	0	0.850	0.059	0.809
1208	obs. noise	Oracle	1	0.700	0.061	0.803
1209	obs. noise	Oracle	2	0.650	0.066	0.808
1210	obs. noise	Oracle	3	0.800	0.059	0.817
1211	obs. noise	Oracle	4	0.700	0.056	0.828
1212	obs. noise	Oracle	5	0.850	0.055	0.819
1213	obs. noise	Oracle	6	0.700	0.058	0.826
1214	obs. noise	Oracle	7	0.750	0.061	0.807
1215	obs. noise	Robust-mean	0	0.650	0.069	0.775
1216	obs. noise	Robust-mean	1	0.500	0.079	0.744
1217	obs. noise	Robust-mean	2	0.400	0.089	0.740
1218	obs. noise	Robust-mean	3	0.500	0.079	0.762
1219	obs. noise	Robust-mean	4	0.650	0.066	0.796
1220	obs. noise	Robust-mean	5	0.600	0.073	0.757
1221	obs. noise	Robust-mean	6	0.400	0.079	0.764
1222	obs. noise	Robust-mean	7	0.450	0.080	0.749
1223	obs. noise	Robust	0	0.650	0.067	0.784
1224	obs. noise	Robust	1	0.500	0.081	0.738
1225	obs. noise	Robust	2	0.250	0.103	0.692
1226	obs. noise	Robust	3	0.600	0.067	0.786
1227	obs. noise	Robust	4	0.600	0.069	0.782
1228	obs. noise	Robust	5	0.450	0.081	0.730
1229	obs. noise	Robust	6	0.550	0.073	0.778
1230	obs. noise	Robust	7	0.650	0.074	0.773
1231	target shift	Adaptive ID	0	0.500	0.087	0.787
1232	target shift	Adaptive ID	1	0.600	0.073	0.819
1233	target shift	Adaptive ID	2	0.350	0.094	0.770
1234	target shift	Adaptive ID	3	0.450	0.081	0.801
1235	target shift	Adaptive ID	4	0.350	0.092	0.771
1236	target shift	Adaptive ID	5	0.450	0.089	0.778
1237	target shift	Adaptive ID	6	0.450	0.087	0.785
1238	target shift	Adaptive ID	7	0.350	0.088	0.777
1239	target shift	Branch v5	0	0.450	0.093	0.774
1240	target shift	Branch v5	1	0.550	0.081	0.804
1241	target shift	Branch v5	2	0.350	0.092	0.776
	target shift	Branch v5	3	0.500	0.079	0.807
	target shift	Branch v5	4	0.400	0.091	0.775
	target shift	Branch v5	5	0.450	0.083	0.791
	target shift	Branch v5	6	0.450	0.090	0.778
	target shift	Branch v5	7	0.450	0.084	0.789
	target shift	Ensemble	0	0.500	0.087	0.787
	target shift	Ensemble	1	0.650	0.070	0.827
	target shift	Ensemble	2	0.350	0.094	0.771
	target shift	Ensemble	3	0.450	0.080	0.804
	target shift	Ensemble	4	0.350	0.094	0.766
	target shift	Ensemble	5	0.500	0.081	0.794

Split	Method	Seed	Success	Final dist.	Progress
target shift	Ensemble	6	0.450	0.089	0.781
target shift	Ensemble	7	0.450	0.083	0.791
target shift	Oracle	0	0.700	0.074	0.818
target shift	Oracle	1	0.700	0.066	0.836
target shift	Oracle	2	0.550	0.079	0.808
target shift	Oracle	3	0.450	0.078	0.807
target shift	Oracle	4	0.450	0.079	0.805
target shift	Oracle	5	0.700	0.067	0.833
target shift	Oracle	6	0.800	0.064	0.842
target shift	Oracle	7	0.500	0.075	0.809
target shift	Robust-mean	0	0.500	0.097	0.763
target shift	Robust-mean	1	0.650	0.075	0.815
target shift	Robust-mean	2	0.350	0.096	0.767
target shift	Robust-mean	3	0.450	0.081	0.801
target shift	Robust-mean	4	0.350	0.097	0.758
target shift	Robust-mean	5	0.500	0.081	0.796
target shift	Robust-mean	6	0.450	0.094	0.767
target shift	Robust-mean	7	0.500	0.083	0.791
target shift	Robust	0	0.600	0.080	0.805
target shift	Robust	1	0.750	0.065	0.841
target shift	Robust	2	0.450	0.086	0.790
target shift	Robust	3	0.450	0.077	0.814
target shift	Robust	4	0.500	0.082	0.794
target shift	Robust	5	0.600	0.068	0.830
target shift	Robust	6	0.500	0.080	0.802
target shift	Robust	7	0.550	0.082	0.793

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